Network Security

Unit	TOPICS/UNITS	Chapter Reference
Unit 1: Introduction to Network Security and Network Fundamentals	• **Intro & Trends:** Importance of security, CIA Triad (Confidentiality, Integrity, Availability), Types of security threats and attacks. • **Network Fundamentals:** OSI and TCP/IP models. IP addressing and subnetting. Networking devices (switches, routers, firewalls). • **Protocols:** Network protocols and services (HTTP, HTTPS, FTP, SSH).	**[1]** Ch 1: 1.1–1.6 (Security Trends); OR **[2]** Ch 1: 1.1 - 1.5, 1.7 **[3]** pg 513–547 (OSI/TCP, Devices,Protocols, IP addressing and subnetting) pg.634 - 636, pg.646 (Networking devices) **[2]** Ch17:17.3 (HTTPS), 17.4(SSH)
Unit 2: Cryptography Basics, Authentication and Secure Design	• **Cryptography:** Symmetric vs Asymmetric encryption, Hash functions, Digital Signatures, PKI, Algorithms (AES, RSA, SHA). • **Authentication & Access:** Passwords, Biometrics, SSO, MFA. Access Control Models (RBAC, ABAC, MAC, DAC). • **Secure Design:** DMZ, VPNs.	**[2]** 6.1 - 6.5 (AES), Ch9:9.2(RSA),Ch 11:11.4(SHA), Ch11(intro), 11.1-11.2 {hash}, Ch 13 (digital sign) **[3]** pg 390-394 {symmetric and asymmetric encryption} **[1]** pg 484 - 487 {PKI} **[2]** pg 506 {MFA} **[3]** pg 832- 841 {Access Control Models }, pg 802-804 {SSO} **[1]** 14.2,14.5 {passwords, biometrics} **[2]** pg 679-680 {DMZ} **[3]** pg 716-724 {VPNs}

Unit 3: Firewalls, IDS/IPS, and Wireless Security	• **Perimeter Security:** types of Firewalls (Packet filtering, Stateful inspection, Application layer), IDS and IPS (concept), Signature and Anomaly based detection, Honeypots and honeynets. • **Wireless:** Standards (802.11, Bluetooth, RFID), and Wireless attacks, securing wireless networks.	**[2]** pg 673 - 679 (types of Firewalls), 681 - 683 (IDS, signature or misuse detection and anomaly detection) **[3]** 873 - 874 (IPS), 1032 - 1033 (honeypots and honeynets) **[3]** pg 570 - 573 (802.11), pg 575 (bluetooth) **[2]** pg 443 - 444 (RFID), pg 568 - 571 (securing wireless network), pg 569 - 570 (Wireless Network Threats or wireless attacks)
Unit 4: Endpoint Security, Malware & Incident Response	• **Endpoint & Malware:** Antivirus and antimalware solutions, Patch management, MDM. Malware types. • **Monitoring & IR:** SIEM systems, Incident Response process, Forensic analysis and evidence handling.	**[3]** pg 1029 - 1031 (patch management) **[2]** pg 686 - 687 (types of malware), pg 687 - 689 (malware defence or antimalware and antivirus solutions), pg 571 - 575 (mobile device management or mobile device security) **[3]** pg 1034 (SIEM), pg 1045 - 1046 (Computer Forensics and Proper Collection of Evidence) [2] pg 688 (Incident Response process or incident management)
Unit 5: Network Security Best Practices and Compliance	• **Governance:** Security policies and procedures, Risk assessment and management, Awareness training. • **Compliance:** Regulatory compliance (HIPAA, PCI-DSS).	**[3]** pg 122 - 125 (Security policies), pg 128 (Security procedures), pg 129 - 131 (Risk management). pg 138-139 (Risk assessment), Pg 194 - 196 (Awareness Training) **[3]** pg 110 - 117 (Regulatory compliance)

References

[1] Behrouz A. Forouzan, Introduction to Cryptography and Network Security. (File: ref_1.pdf)
[2] William Stallings, Cryptography and Network Security: Principles and Practice, 8th Edition. (File: ref_2.pdf)
[3] Shon Harris, Fernando Maymí, CISSP All-in-One Exam Guide, 8th Edition. (File: ref_3.pdf)
[4] Atul Kahate, Cryptography and Network Security, 2nd Edition. (File: ref_4.pdf)